cutting problems

digitalisierung und programmierung · prof. dr. nicolas meseth

1. you can already do this
2. the project's elephant
3. three ways to cut
4. a good cut is testable
5. cuts are agreements

part 1

you can already do this

how do you eat an elephant?



image: ai-generated (gpt-image-2)

you plan like this without thinking

party, saturday, 30 guests

you plan like this without thinking

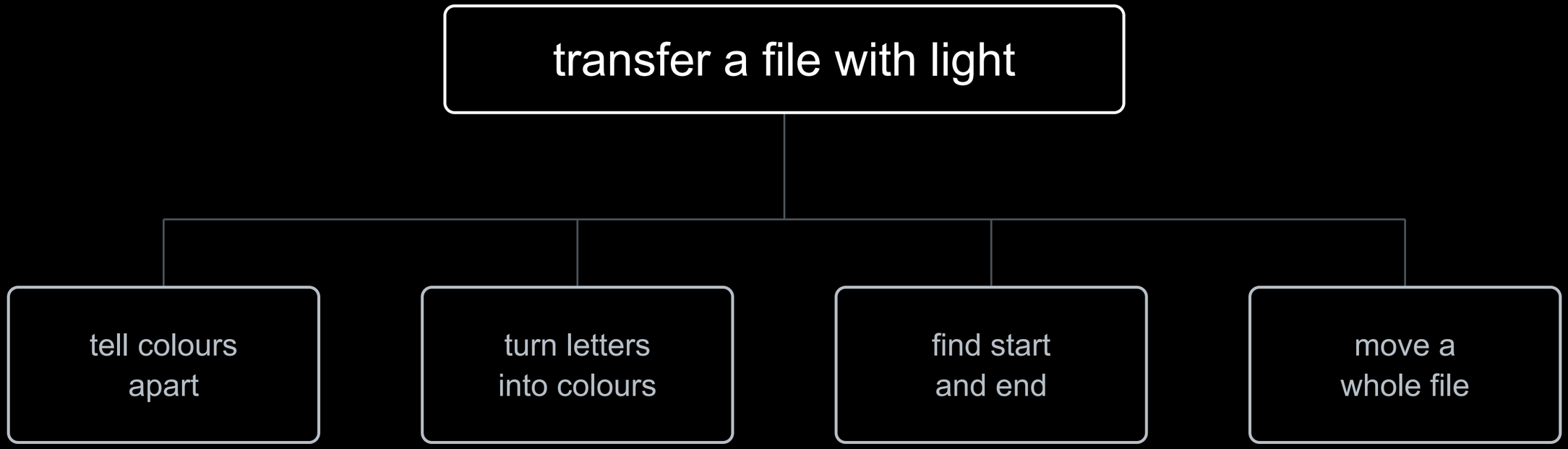


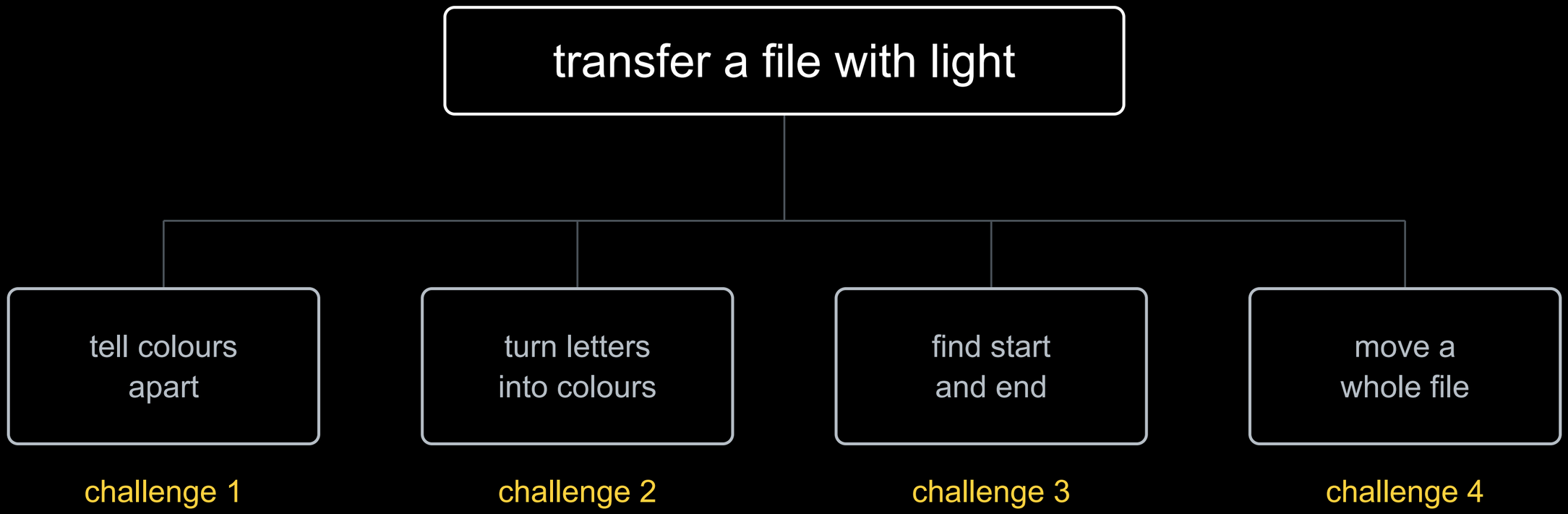
part 2

the project's elephant

transfer a file from one laptop
to another, using light.

transfer a file with light


transfer a file with light
tell colours apart
turn letters into colours
find start and end
move a whole file


transfer a file with light
tell colours apart
turn letters into colours
find start and end
move a whole file
challenge 1
challenge 2
challenge 3
challenge 4

how do computers solve complex problems?

the skills for it: cutting problems · input, processing, output
algorithms and programs · measuring · thinking in layers



colours into symbols, letters into bits

a file is bytes; what holds them?

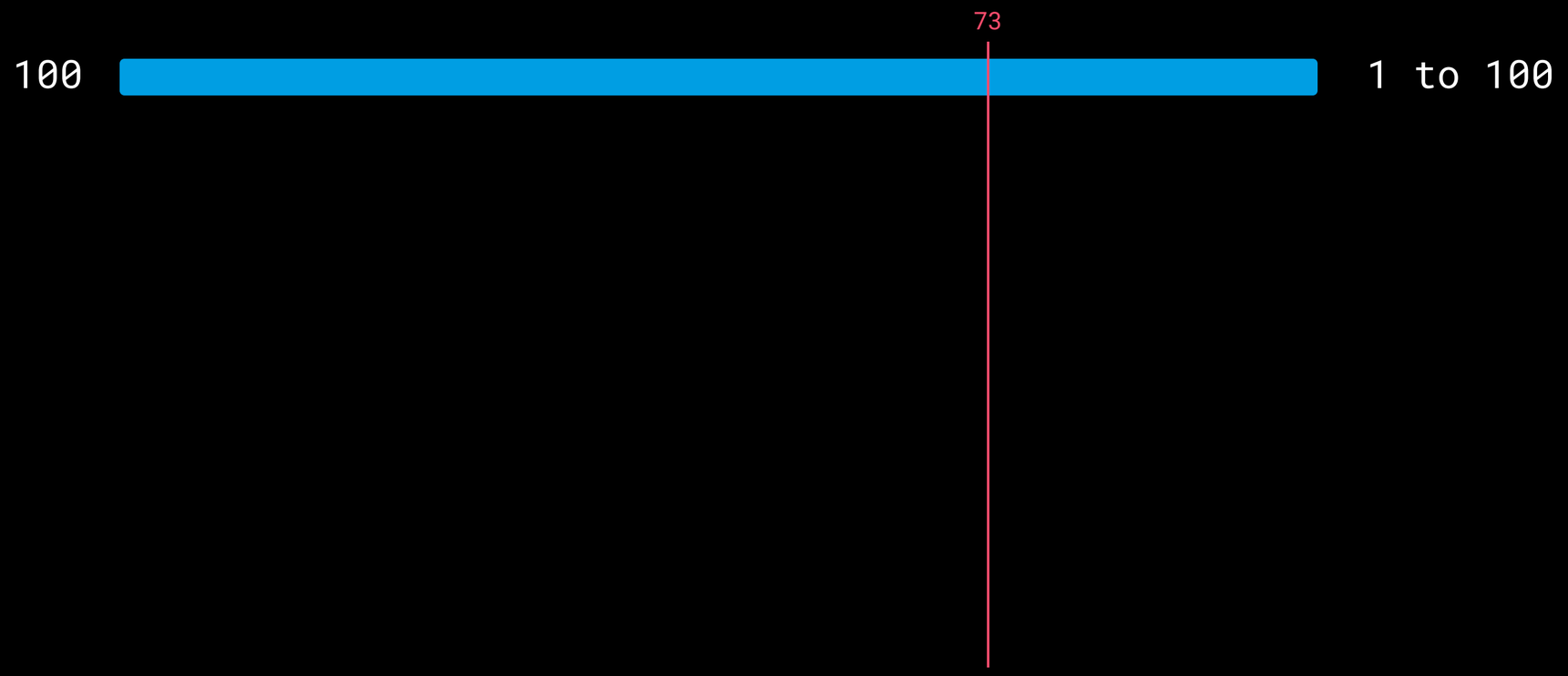
signal and noise, timing, protocols

checksums, filters, encryption

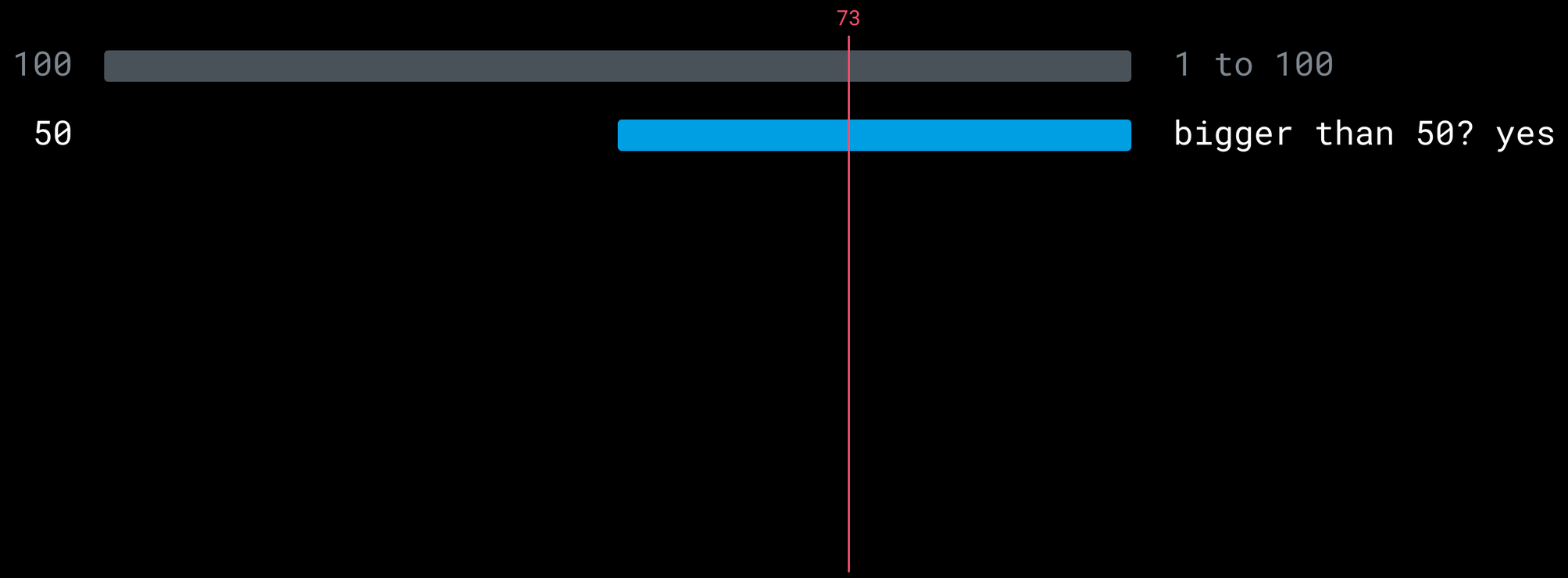
part 3

three ways to cut

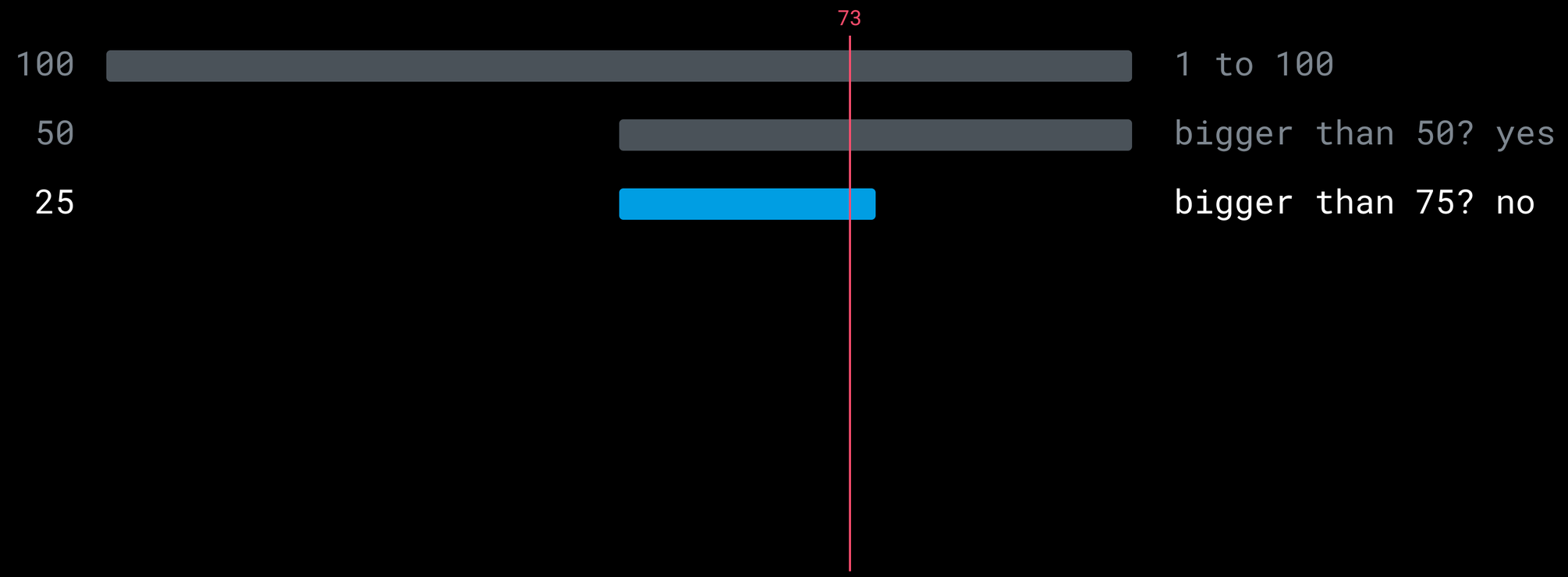
divide & conquer: halve until it's easy



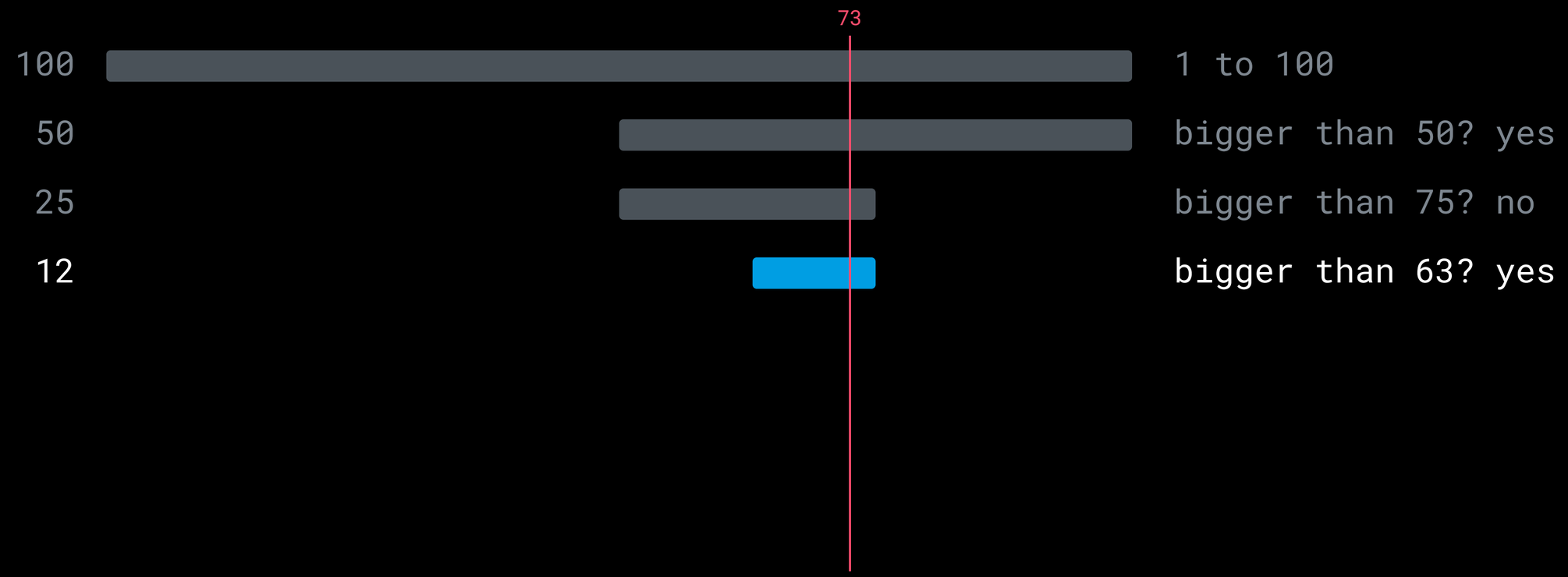
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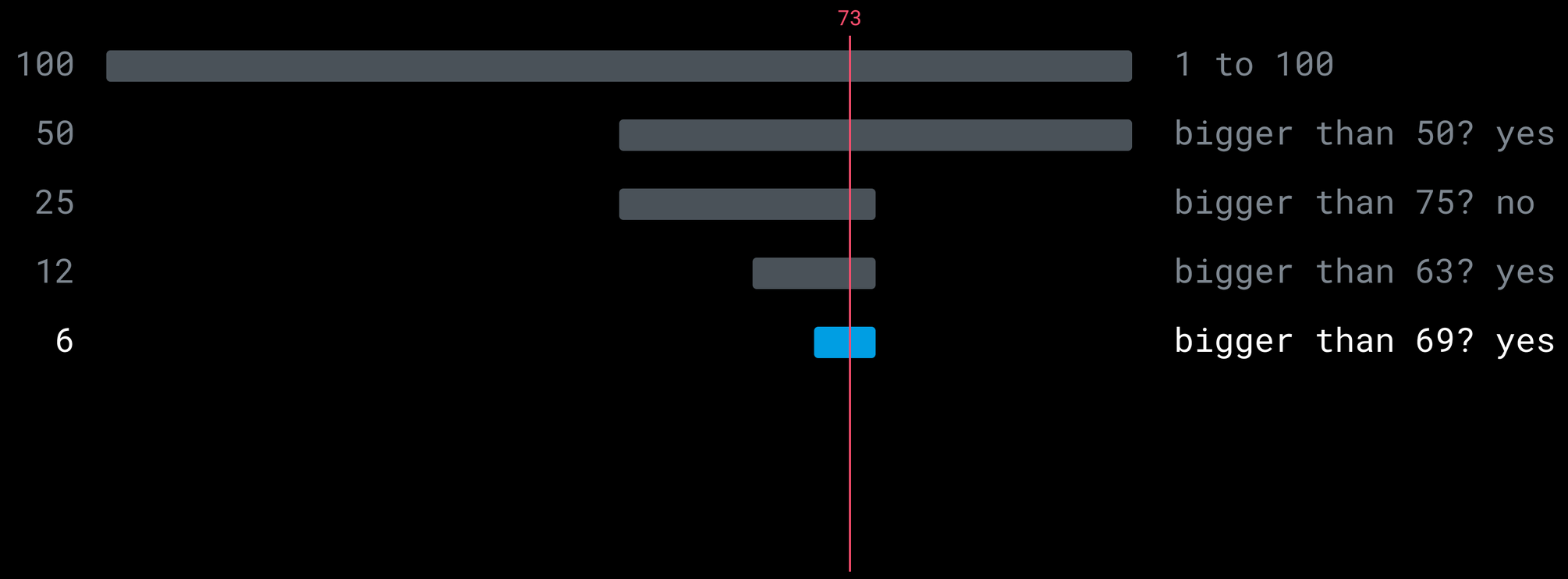
divide & conquer: halve until it's easy



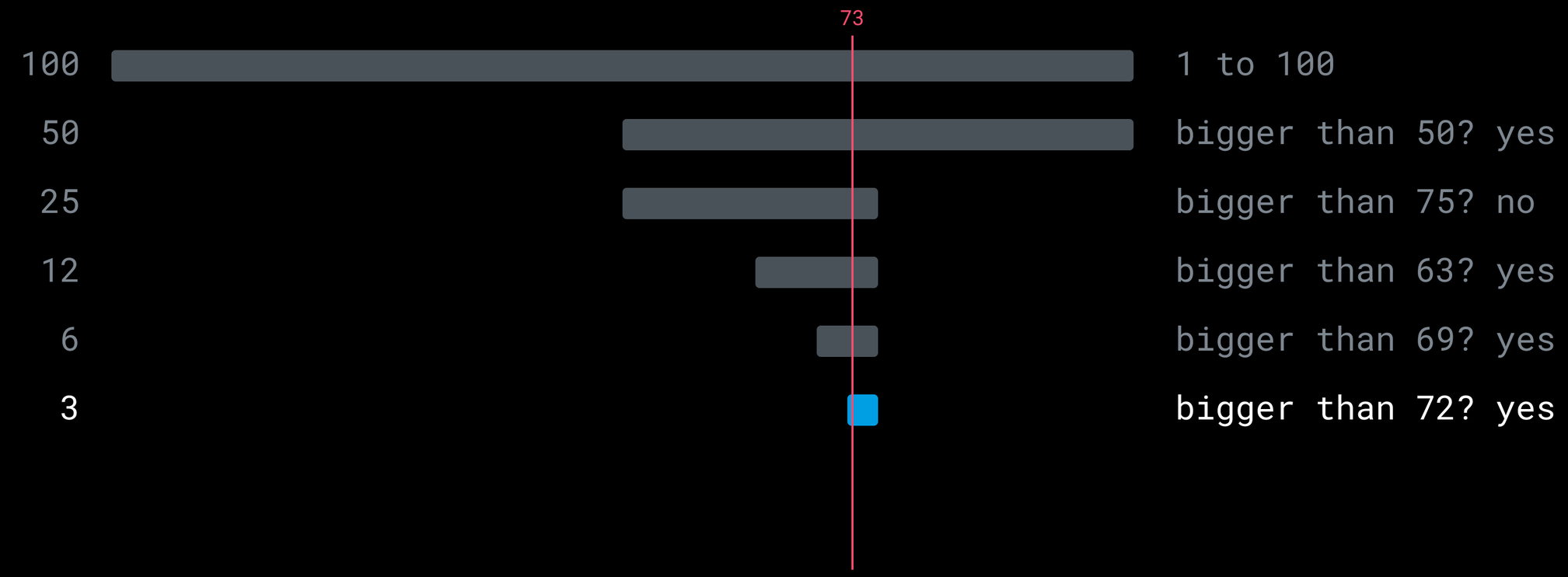
divide & conquer: halve until it's easy



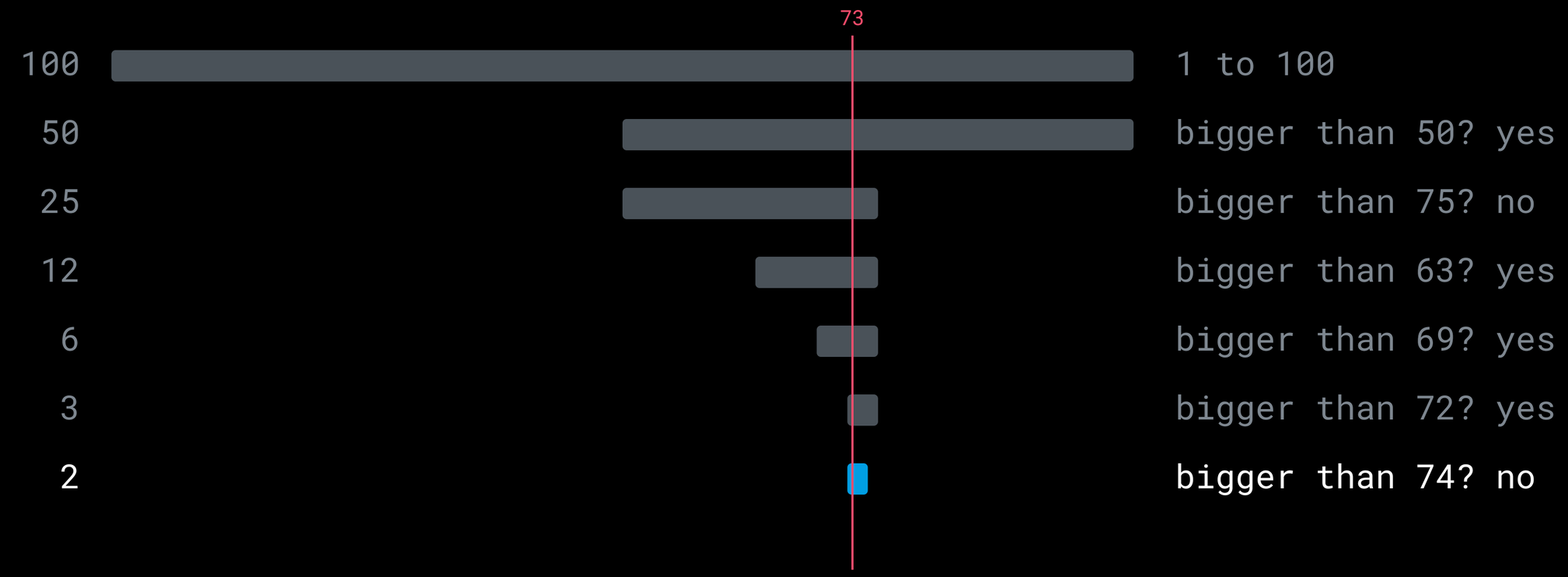
divide & conquer: halve until it's easy



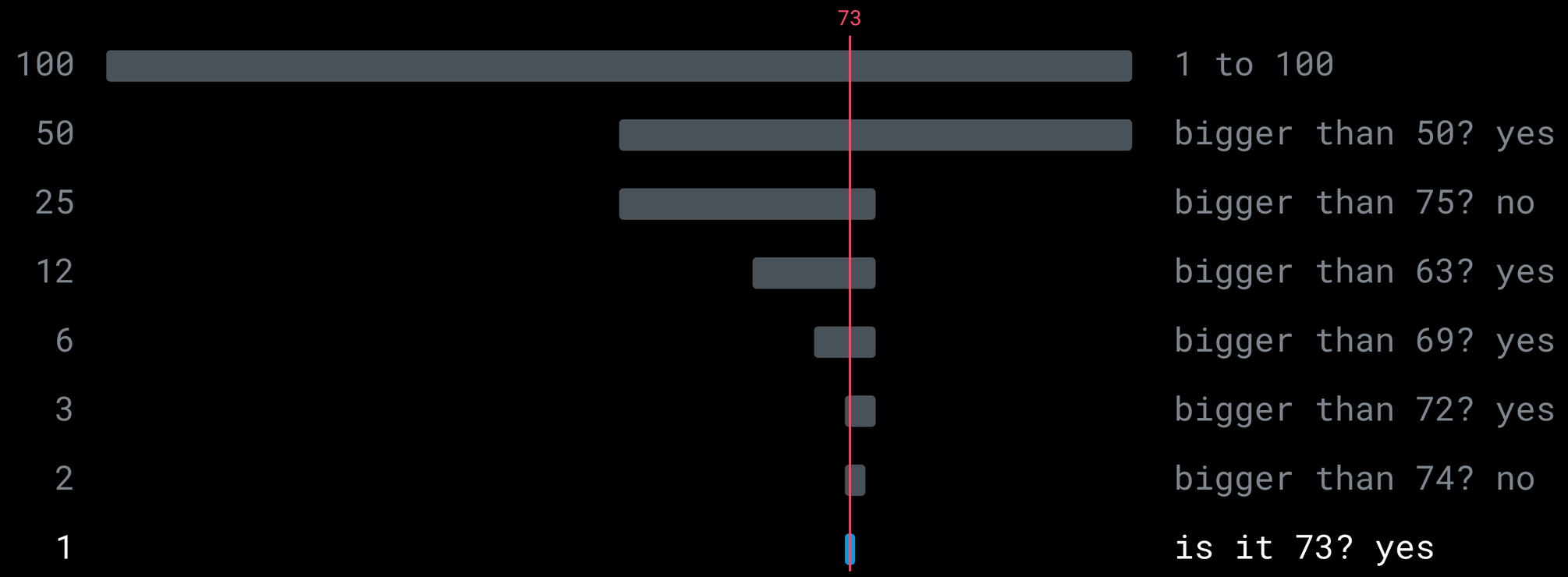
divide & conquer: halve until it's easy



divide & conquer: halve until it's easy



divide & conquer: halve until it's easy



seven questions are enough, whatever the number.

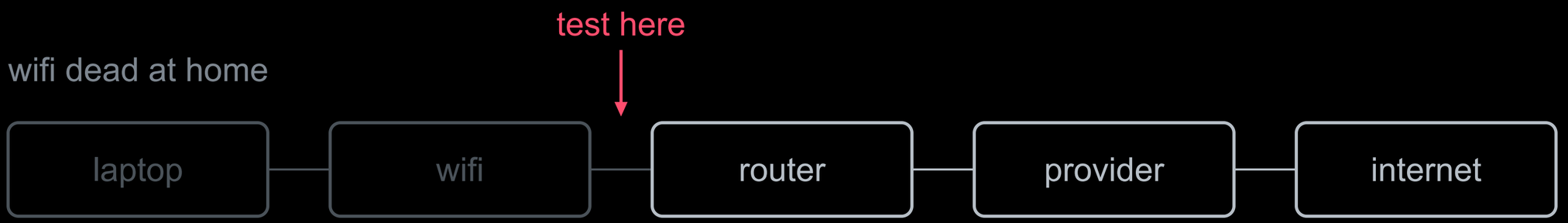
how many steps in general?

the same trick when something breaks

wifi dead at home

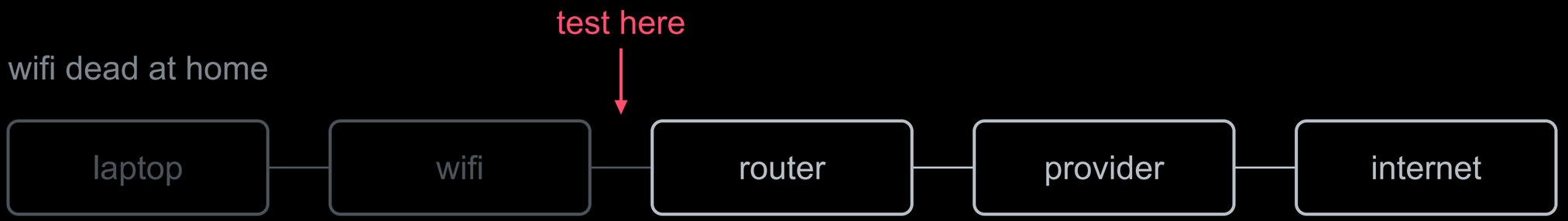


the same trick when something breaks

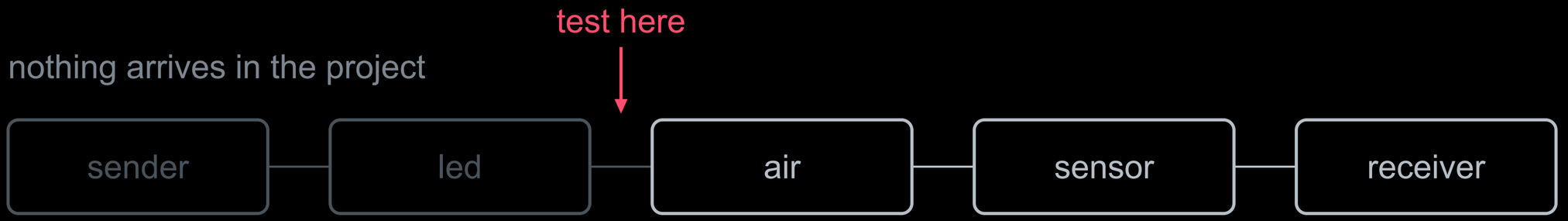


the same trick when something breaks

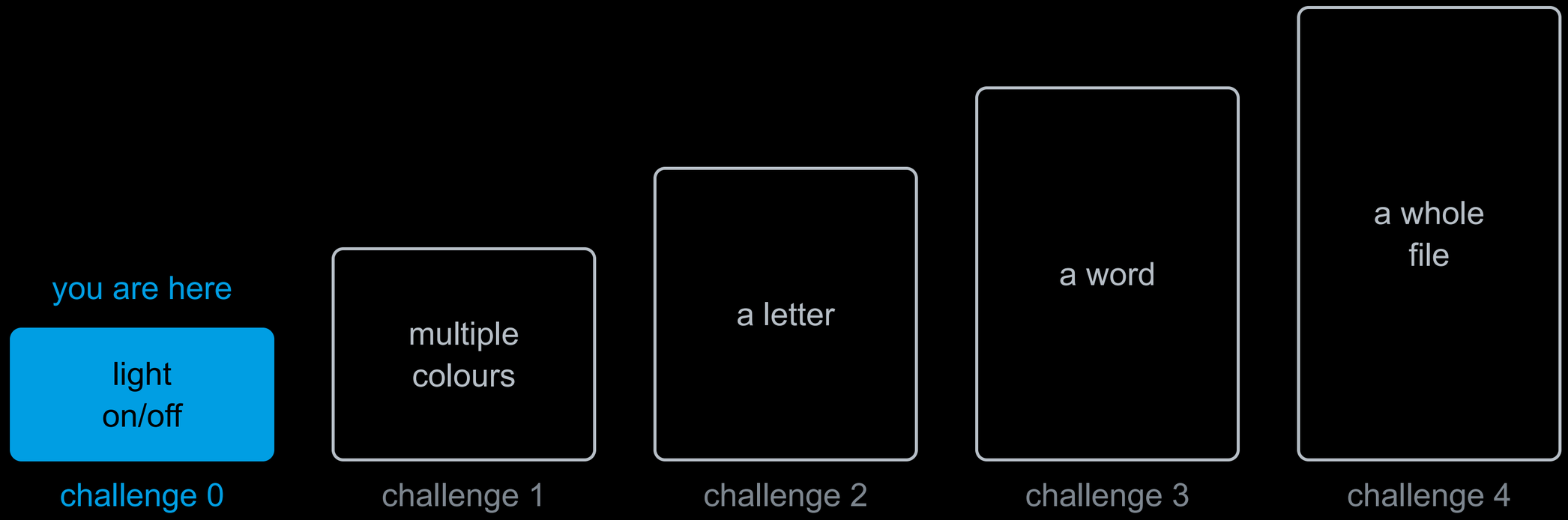
wifi dead at home



nothing arrives in the project



start smaller: solve an easier version first



part 4

a good cut is testable

which one are you actually preparing?

i need to learn maths.

i can pass the 2023 exam in 90 minutes.

which one are you actually preparing?

i need to learn maths.

i can pass the 2023 exam
in 90 minutes.



the led works.

red or blue?
right 50 times in a row.

the led works.

red or blue?
right 50 times in a row.

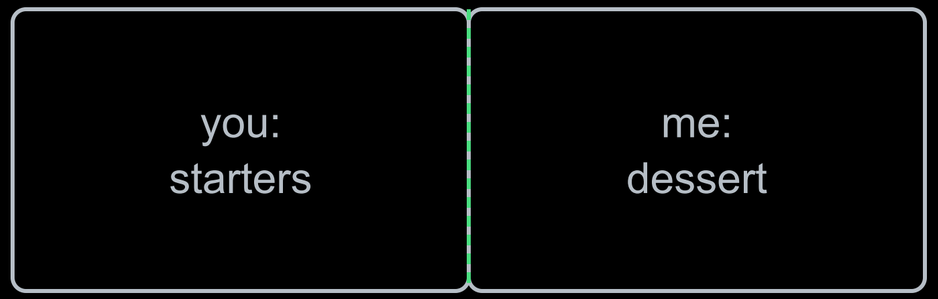


done means: the test passes.

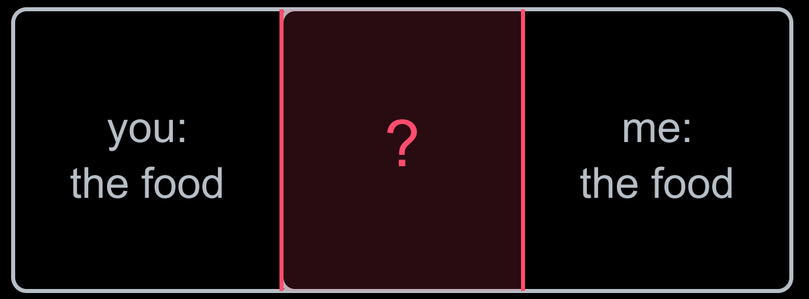
part 5

cuts are agreements

where you cut decides how much you must talk



works: the border is clear



chaos: who buys the bread?

and this is how you talk to the AI assistant

one big wish

```
build me the
file transfer.
```

one small piece, with its test

```
read the sensor ten times
and return the average.
test: covered, it stays under 10.
red led on, it goes over 100.
```

you cut. the assistant solves pieces. you check.

and this is how you talk to the AI assistant

one big wish

```
build me the
file transfer.
```

one small piece, with its test

```
read the sensor ten times
and return the average.
test: covered, it stays under 10.
red led on, it goes over 100.
```



you cut. the assistant solves pieces. you check.

one bite
at a time.



image: ai-generated (gpt-image-2)